Named Entity Recognition with spaCy

Using Hugging Face Transformers to do Named Entity Recognition

Kabui, Charles

2025-05-03

Table of Contents

Read at ToKnow.ai | Open in Kaggle | Download as Notebook

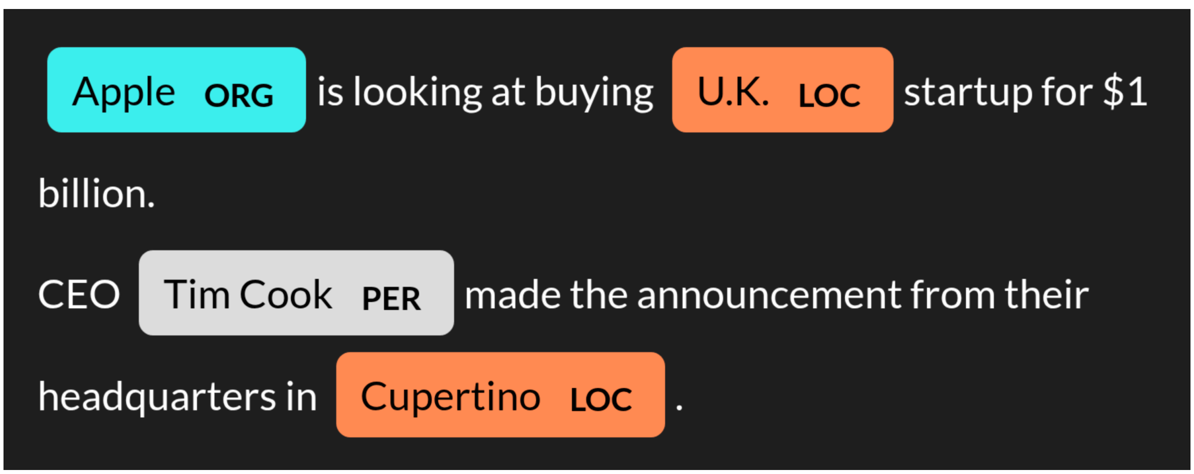

Figure 1: Named-entity recognition (NER) Example

Named-entity recognition (NER) is about identifying and classifying named entities in text into predefined categories such as `persons`, `organizations`, `locations`, etc. It extracts entities from unstructured text.

By identifying and categorizing real-world entities in text, NER bridges the gap between unstructured text and structured information that can drive decision-making and insights. Here's why it's so important:

1. Business Intelligence & Market Analysis
    - **Competitive Intelligence**: Extract competitor names, products, and initiatives from news articles and reports
    - **Market Trends**: Identify emerging players, technologies, and market shifts by analyzing mentions of organizations and products
    - **Customer Insights**: Extract brand mentions, product names, and sentiment indicators from customer feedback
2. Information Extraction & Knowledge Management
    - **Automated Data Population**: Populate databases and knowledge graphs with structured information extracted from unstructured text
    - **Document Classification**: Enhance document categorization by understanding the entities mentioned within content
    - **Content Recommendation**: Improve content recommendations by matching based on entities of interest to users
3. Search & Retrieval Enhancement
    - **Semantic Search**: Enable entity-based search beyond simple keyword matching
    - **Question Answering**: Power systems that can answer "who", "where", and "when" questions by extracting relevant entities
    - **Document Summarization**: Identify key figures, organizations, and locations for inclusion in automated summaries
4. Healthcare & Biomedical Applications
    - **Clinical Data Mining**: Extract patient information, conditions, treatments, and medications from clinical notes
    - **Medical Literature Analysis**: Identify relationships between diseases, drugs, and genes in research papers
    - **Clinical Trial Matching**: Match patients to appropriate clinical trials based on extracted medical entities
5. Finance & Risk Management
    - **Regulatory Compliance**: Extract reporting entities and requirements from regulatory documents
    - **Financial News Analysis**: Track mentions of companies, executives, and financial metrics in real-time
    - **Risk Assessment**: Identify potentially problematic entities (people, organizations) for due diligence

6. Legal Document Analysis

    - **Contract Analysis**: Extract parties, dates, monetary values, and conditions from legal documents
    - **Case Law Research**: Identify precedents, judges, courts, and legal principles in legal opinions
    - **Compliance Monitoring**: Extract regulated entities and requirements from regulatory documents

7. Security & Intelligence

    - **Threat Detection**: Identify potentially suspicious individuals, organizations, or locations in communications
    - **Network Analysis**: Build relationship graphs between entities mentioned in various documents
    - **Predictive Analysis**: Use historical patterns of entity mentions to predict future events or trends

8. Benefits Across Applications

    - **Automation**: Reduce manual data entry and information extraction
    - **Consistency**: Apply uniform entity extraction rules across large document collections
    - **Discovery**: Uncover hidden relationships between entities across disparate sources
    - **Time Efficiency**: Process volumes of text at scale that would be impossible manually
    - **Structured Data Creation**: Transform unstructured text into structured, queryable information

Example

```
from transformers import AutoTokenizer, AutoModelForTokenClassification, pipeline
import spacy
from spacy import displacy
```

Initialize the NER processor with a transformer model using the (`dslim/bert-base-NER`).

Alternative model names for NER

- "jean-baptiste/roberta-large-ner-english"
- "dbmdz/bert-large-cased-finetuned-conll03-english"
- "elastic/distilbert-base-cased-finetuned-conll03-english"

For domain-specific NER, look for models trained on data from your domain.

For example, "emilyalsentzer/Bio_ClinicalBERT" for medical text.

```
model_name="dslim/bert-base-NER"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForTokenClassification.from_pretrained(model_name)
ner_pipeline = pipeline(
    "ner",
    model=model,
    tokenizer=tokenizer,
    aggregation_strategy="simple")
```

```
Some weights of the model checkpoint at dslim/bert-base-NER were not used when initializing B
- This IS expected if you are initializing BertForTokenClassification from the checkpoint of
- This IS NOT expected if you are initializing BertForTokenClassification from the checkpoint
Device set to use cpu
```

Extract named entities from text

```
text = """Apple is looking at buying U.K. startup for $1 billion.
CEO Tim Cook made the announcement from their headquarters in Cupertino.
"""
entities_list = ner_pipeline(text)
entities_list
```

```
[{'entity_group': 'ORG',
  'score': np.float32(0.9989743),
  'word': 'Apple',
  'start': 0,
  'end': 5},
 {'entity_group': 'LOC',
  'score': np.float32(0.98070335),
  'word': 'U. K.',
  'start': 27,
  'end': 31},
 {'entity_group': 'PER',
  'score': np.float32(0.9997685),
  'word': 'Tim Cook',
  'start': 61,
  'end': 69},
```

```
 {'entity_group': 'LOC',
  'score': np.float32(0.9980777),
  'word': 'Cupertino',
  'start': 119,
  'end': 128}]
```

Visualize entities in text using spaCy's displacy.

```
nlp = spacy.blank("en")
# Create a spaCy Doc
doc = nlp(text)

# Build spaCy Span objects for each entity
spans = [
    doc.char_span(
        start_idx = ent["start"],
        end_idx = ent["end"],
        label = ent["entity_group"],
        alignment_mode = "contract")
    for ent
    in entities_list]


# Set the entities in the Doc
doc.ents = filter(lambda x:x != None, spans)


# Display using displacy
displacy.render(doc, style="ent", jupyter=True)
```

Please visit https://toknow.ai/posts/named-entity-recognition/index.html to view the output of this code cell.
